

Agents Unite: Crowdsourced LLM Agents as a Git-Native Financial Memory

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Large language model (LLM) agents can synthesize market narratives at scale, but solo deployments waste tokens, discard history, and miss timezone-specific information. We present *Agents Unite*, an architecture that treats GitHub as a collective memory substrate for distributed financial intelligence. Each participant runs a scheduled local agent that researches one deterministically assigned ticker per day, submits a structured report as a pull request, and optionally serves as a verifier or consensus writer. Raft-style leader election resolves hourly write conflicts; reputation-weighted aggregation merges independent reports into a dated ledger. The design applies collective intelligence ideas—task decomposition, diversity, and consensus—without blockchain infrastructure: Git commits, CI validation, and CODEOWNERS provide auditability. Per-operator inference cost drops from hundreds of dollars per day for solo full-universe coverage to roughly five cents per contributor per day; readers fork the archive at zero marginal inference cost. This poster presents an early proof-of-concept architecture and invites feedback on assignment, verification, and herding mitigation before a public benchmark evaluation.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing systems and tools**; • **Computing methodologies** → *Multi-agent systems*; • **Applied computing** → *Economics*.

Additional Key Words and Phrases: collective intelligence, human-AI collaboration, multi-agent systems, crowdsourcing, financial data

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1 Motivation and Collective Intelligence Framing

Financial markets generate narrative, sentiment, and price information faster than any single analyst or agent can absorb. Recent LLM agents help, but three failures persist in solo deployments: *ephemeral memory* (yesterday’s analysis must be regenerated at full cost), *budget exhaustion* (covering thousands of tickers daily costs hundreds to thousands of dollars in API tokens), and *temporal blind spots* (a single timezone misses overnight filings and foreign-market chatter).

Agents Unite reframes market research as a **crowd intelligence** problem rather than a single-agent scaling problem [1]. Instead of one operator researching the entire universe, many contributors each research one ticker per cycle. Assignment is deterministic—contributors cannot cherry-pick popular symbols—and roles (research, verify, consensus) are assigned at run time by a scheduled local runner. The system is designed for human-AI complementarity: humans may contribute manually or run agents with different model providers (hosted APIs, local models, or coding-agent adapters), while CI and peer verifiers provide lightweight quality control.

This work is affiliated with the **collective intelligence (CI)** track: it studies how decentralized agents, diverse information sources, and consensus mechanisms combine into a longitudinal memory that no individual node could economically maintain alone.

2 System Design

Figure 1 summarizes the pipeline. There is no central application server; GitHub is the database.

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Assign → Research → Validate → PR → CI → Verify → Consensus → data/DATE/TICKER/

Fig. 1. End-to-end ingestion: local agents produce reports; GitHub CI and peer verifiers gate merges.

Deterministic assignment. Ticker T_k is chosen by hashing the date and contributor identity modulo universe size $|T|$. A parallel lottery assigns roles, adjusted by a CI-generated coverage manifest that prioritizes under-covered sectors, geographies, and modalities.

Verification without expensive re-research. Verifiers inspect public pull requests using smaller models and deterministic consistency checks on cited sources, escalating uncertain claims to yes/no or cross-encoder natural language inference checks. Participants do not pre-declare as verifiers; the schedule decides.

Hourly leader election. For Phase 3 hourly shards, multiple pull requests may arrive for the same (date, ticker, hour). After the hour window closes, CI recomputes

$$\text{leader} = \arg \min_i \text{SHA256}(\text{date} : \text{hour} : \text{ticker} : \text{hash}_i)$$

among schema-valid submissions. Only the leader merges into the canonical path. Provenance is recorded in the pull request, merge commit, and per-ticker metadata.

Consensus. Validated sentiment scores pass through median aggregation with median absolute deviation outlier rejection, then a reputation-weighted median. Source-diversity weighting is proposed to mitigate herding when contributors cite the same wire story [2].

3 Economics and Early Evidence

Table 1 contrasts solo regeneration with crowd contribution. The gain is per-operator burden reduction: aggregate work still happens, but it is spread across contributors and reused by readers who fork the ledger.

Table 1. Illustrative per-operator annual costs (July 2026 API list rates).

Path	Cost
Solo full-universe (budget)	~\$73k/yr
Solo full-universe (mid)	~\$219k/yr
Agents Unite contributor	~\$18/yr
Agents Unite reader	\$0 inference

External work on federated inference-tree sharing suggests pooled agent memory can sharply outperform single-node caches at fixed false-accept rates [3]; Agents Unite aims to test an analogous effect on ticker-day financial queries once a public benchmark exists. Empirical validation—sentiment factor backtests, ledger reuse, verification cost, and adversarial probes—is planned for a follow-on paper after benchmark release.

4 Poster Contribution and Discussion Questions

This poster presents an **early architecture** for turning fragmented agent labor into a compounding public memory. We seek feedback on four open design choices:

- (1) Whether deterministic assignment plus source-diversity weighting is sufficient to limit herding [2].

- (2) How much verification can be delegated to cheap binary checks versus full agentic re-research.
- (3) Whether Git-native coordination is adequate at multi-thousand-report-per-day scale without hybrid columnar snapshots.
- (4) What governance model, if any, should steward a cross-asset narrative ledger as participation grows.

Outputs are structured observations, not investment advice. A companion architecture paper and conference poster provide additional detail; a live demo repository is under active development.

Acknowledgments

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References

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- [2] Zhi Da and Xing Huang. 2020. Harnessing the Wisdom of Crowds. *Management Science* 66, 5 (2020), 1847–1867. doi:10.1287/mnsc.2019.3294
- [3] Folklore Contributors. 2026. Folklore: Federated Inference-Tree Sharing for Agent Memory. github.com/usefolklore/folklore. BEIR recall@1 at $\leq 2\%$ false-accept: SciFact 81.5% $\rightarrow$ 98.2%, NFCorpus 77.3% $\rightarrow$ 97.6%, FiQA 66.0% $\rightarrow$ 94.5% under federated tree-sharing vs. single-node semantic cache.